Engineering for High Availability and Zero Downtime

Resilience in Business Critical Systems

William Cochrane | 1st November 2023

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YEAR OUT **INDUSTRY PLACEMENTS**.

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Categorising Risk in Critical Systems

Physical Impact

Safety Critical

Risk: Loss of Life / Environmental Damage.

Example: Chemical manufacturing plant, unmanned train metro systems, aviation.

Mission Critical

Risk: Project / Objective Failure.

Example: Spacecraft navigational systems, airport baggage handling.

Digital Impact

Business Critical

Risk: Economic / Reputational Loss.

Example: Bank accounting systems, stock-trading platforms, ERP systems.

Security Critical

Risk: Data Theft.

Example: Loss of sensitive personal or corporate data.

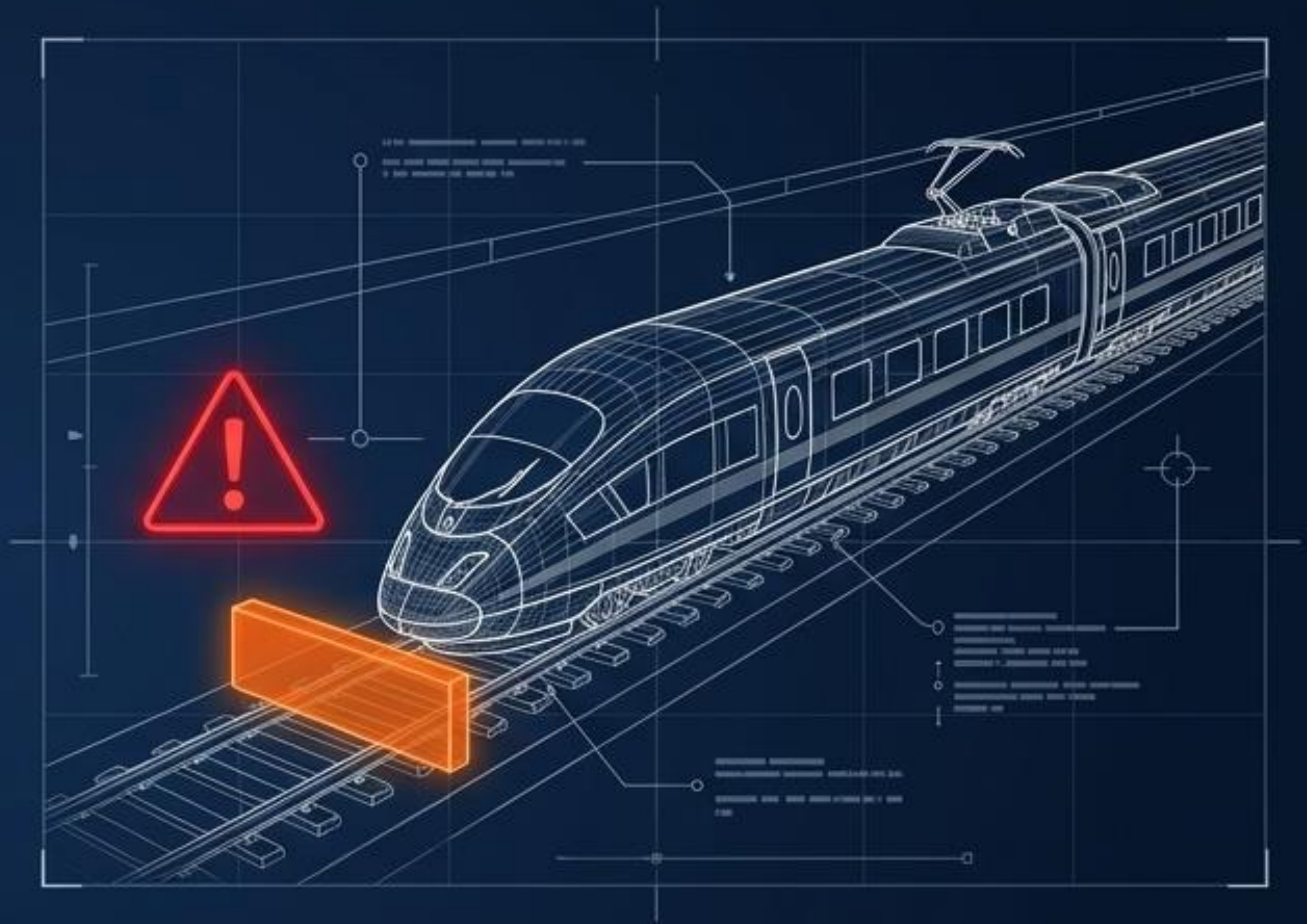
Localized Risk

Systemic Risk

A critical system must remain highly reliable as it evolves, without incurring prohibitive costs.

Two Philosophies for Handling System Failure

Fail-Operational: The system must operate not only in nominal conditions but also in **degraded situations** when parts malfunction.
(e.g., Airplanes must fly even with component failure).



Fail-Safe: The system must safely **shut down** entirely in the event of **single or multiple failures** to **prevent catastrophe**.
(e.g., Stopping a train puts it into a safe state).

The Fallacy of "Good Code"

Write Good Software → Test Thoroughly → Fix Bugs → ~~JOB DONE?~~

The Myth

Software reliability is the same thing as system resilience.

The Reality

Flawless code cannot prevent a router from burning out, a power grid from failing, or an ISP from dropping connection. **Resilience is an infrastructure problem**, not just a software problem.

Defining True System Resilience

1. Continuous running capabilities.
2. Constant accessibility for end-users.
3. High responsiveness under heavy load.
4. Uninterrupted access to dependent data/state.
5. Zero-downtime during system updates.
6. Seamless scalability.

The 6 Pillars of a Resilient Application

Foundational Definition:
“The ability to adapt to changing conditions and withstand and rapidly recover from disruption due to emergencies.” — US Department of Homeland Security.

The Enemy of Availability is the SPOF

Single Point of Failure (SPOF): Any non-redundant part of a system that, if dysfunctional, causes the entire system to fail. It is the direct antithesis of High Availability.

High Availability (HA): An infrastructure configured to operate at a high level continuously, handling different loads and failures with minimal or zero downtime.

The Three Layers of IT Infrastructure

Layer 1: Compute Resource

The active processing layer. Servers, Virtual Machines, and PCs where the application logic actually runs.

Layer 2: Connectivity

The vascular system. Network switches, routers, and load balancers connecting compute resources to each other and the outside world.

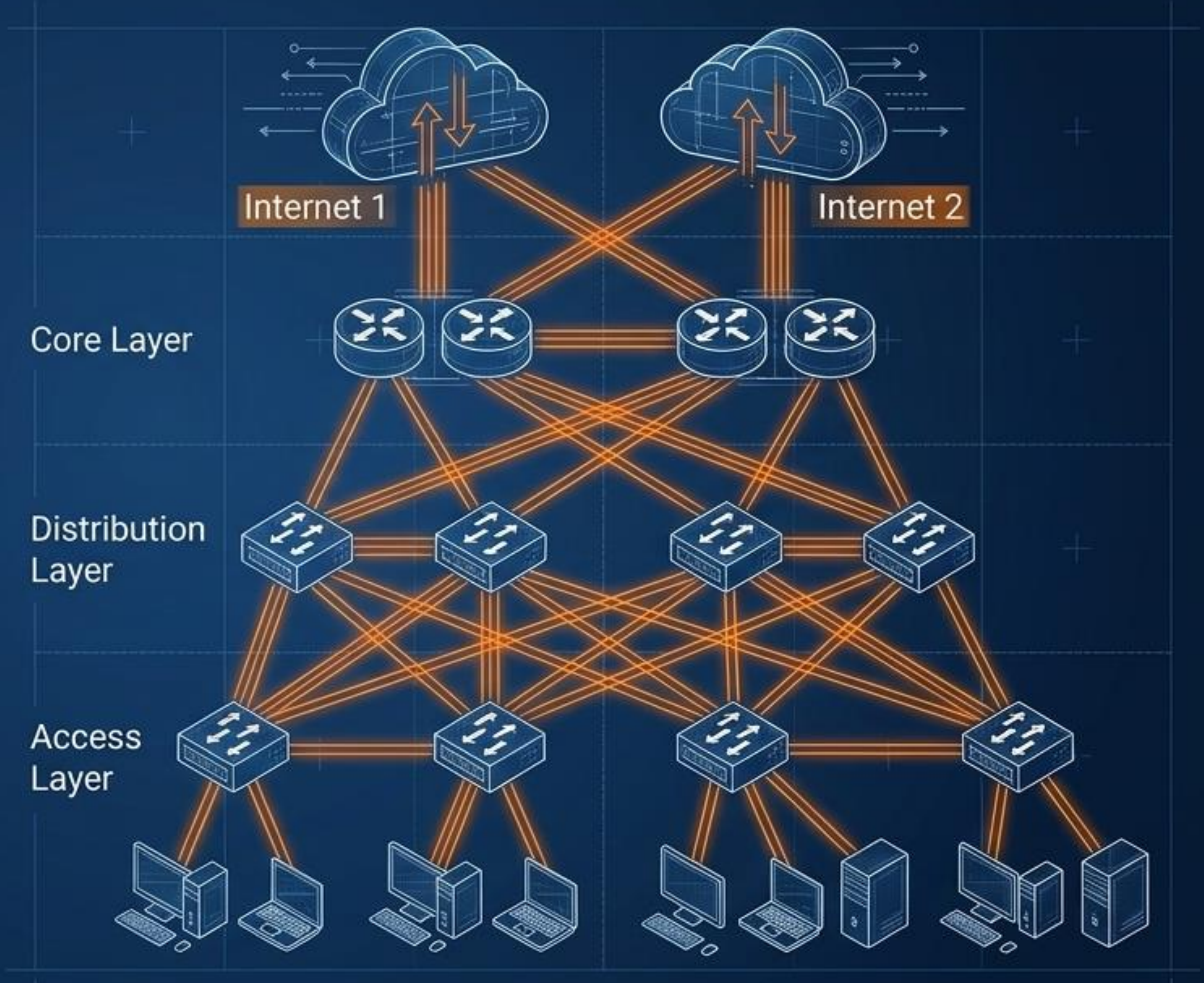
Layer 3: Data Storage

The foundational memory. Persistent storage arrays, NAS file servers, and physical disk drives where state is maintained.

Eradicating Network Single Points of Failure

Redundancy at Every Node:
A resilient network demands multiple overlapping pathways.

Key Defensive Mechanisms:
- Multiple L2/L3 Switches & Load Balancers.
- Multipath connections across Multiple ISPs.
- Dynamic range switching & Network Bonding.
- Multiple Routers utilizing failover protocols (BGP, OSPF, EIGRP, RIP).



Routing Redundancy: Hot-Standby vs. Active-Standby

Hot-Standby

Primary router is active; secondary is offline until failover.

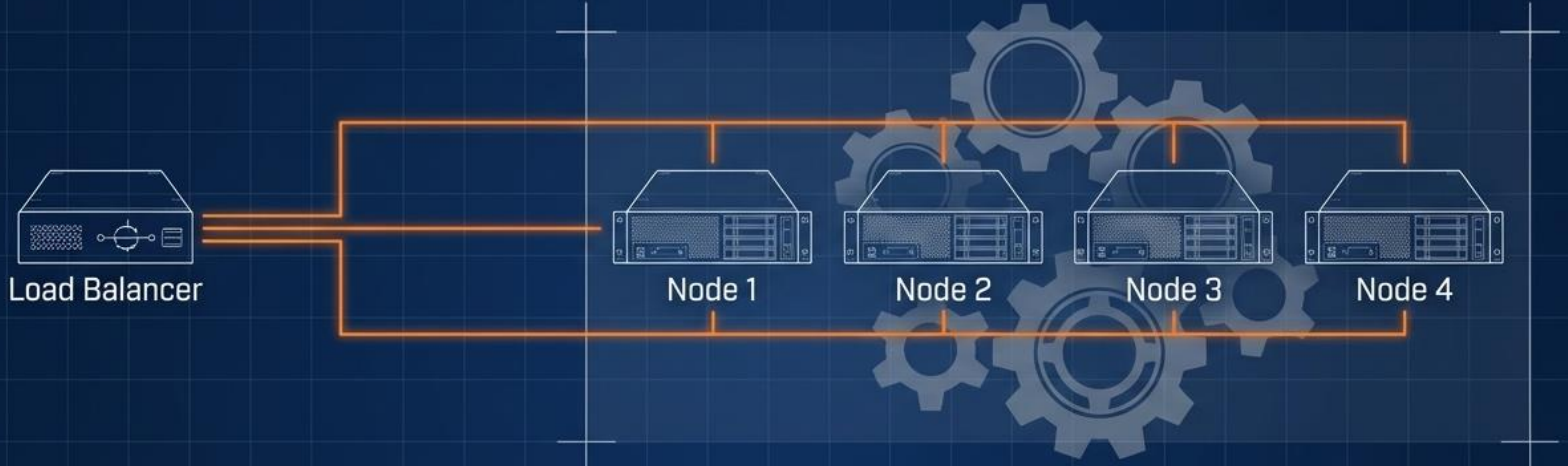
- **The Flaw:** Because they share the same physical interface via a single switch, a hardware failure at the switch level takes down both routers.

Active-Standby

Uses entirely different WAN interfaces on each router, connecting to different ISPs.

- **The Advantage:** Eliminates shared physical hardware, removing the local SPOF and mitigating upstream ISP failures.

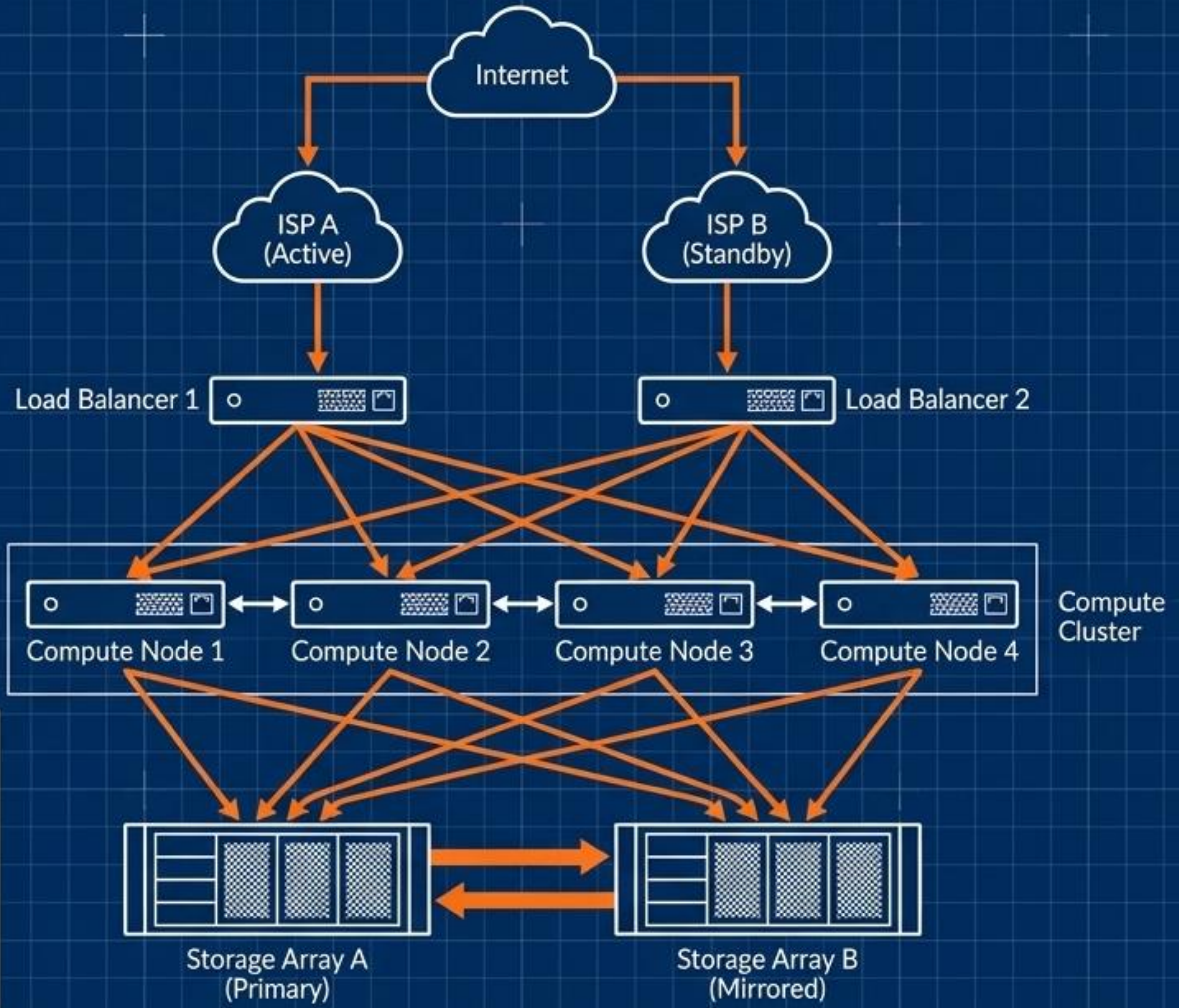
Eradicating Compute SPOF via Server Clusters



The High Availability Cluster:
Instead of relying on one massive server, compute tasks are distributed across multiple individual servers, called Nodes.

Cluster Mechanics:
- Nodes act as a single, unified system under one IP address.
- Workload is dynamically distributed to prevent overload.
- Nodes communicate over a dedicated network to constantly synchronize data.
- If one node dies, traffic seamlessly reroutes to surviving nodes with zero downtime.

The Blueprint of a Resilient Architecture



Holistic Resilience: True high availability is achieved only when the Compute, Connectivity, and Storage layers are all fortified against failure.

Zero SPOFs: In this architecture, any single cable can be cut, or any single server unplugged, and the user experiences zero interruption.

Architect for Inevitable Failure

Mindset Shift

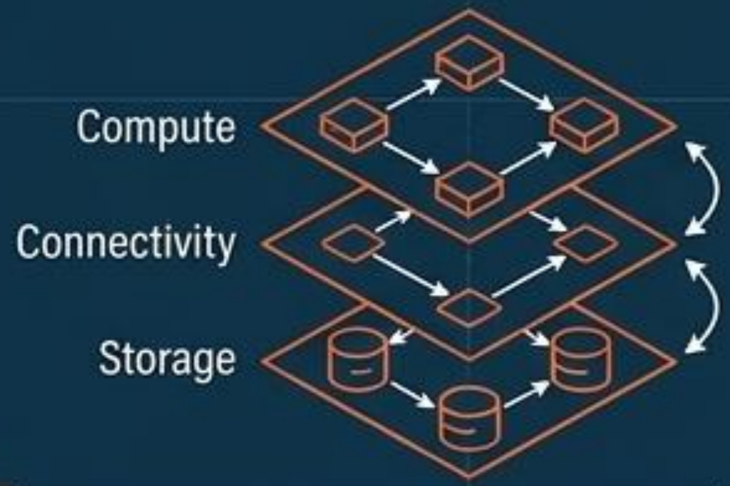
Flawless software running on fragile infrastructure will still fail.

Hunt the SPOF

Ruthlessly identify and engineer around non-redundant bottlenecks.

Layer Defenses

Redundancy must exist across Compute, Connectivity, and Storage simultaneously.



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